

# **SHOPIFY STOCK MARKET ANALYSIS USING POWER BI**

## **WEEK 4 – EXPLORATORY DATA ANALYSIS (EDA)**

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# 1. Introduction

This project focuses on performing Exploratory Data Analysis (EDA) on Shopify stock data using Microsoft Power BI. The analysis helps to understand stock price and trading volume trends through interactive charts, moving averages, slicers and tooltips.

## 2. Objectives

- To analyse Shopify stock price and trading volume.
- To calculate 20-day and 50-day Simple Moving Averages.
- To identify moving average crossover signals.
- To create interactive stock market visualizations.
- To analyse trading volume trends.
- To provide detailed information using custom tooltips.

## 3. Tools Used

- Microsoft Power BI
- Power Query
- DAX (Data Analysis Expressions)
- Shopify Stock Dataset

## 4. Dataset Description

The Shopify stock dataset contains stock market information such as Date, Open Price, High Price, Low Price, Close Price and Volume. This data is used to analyse stock price movements and trading activity.

## 5. Simple Moving Average (SMA)

Simple Moving Average is used to calculate the average value over a specific number of days.

In this project:

- 20-Day SMA is calculated using the previous 20 days of volume data.
- 50-Day SMA is calculated using the previous 50 days of volume data.

DAX measures were created using `DATESINPERIOD()` and `AVERAGE()` functions.

## 6. Moving Average Crossover Signal

A Moving Average Crossover Signal was created by comparing the 20-day SMA and 50-day SMA.

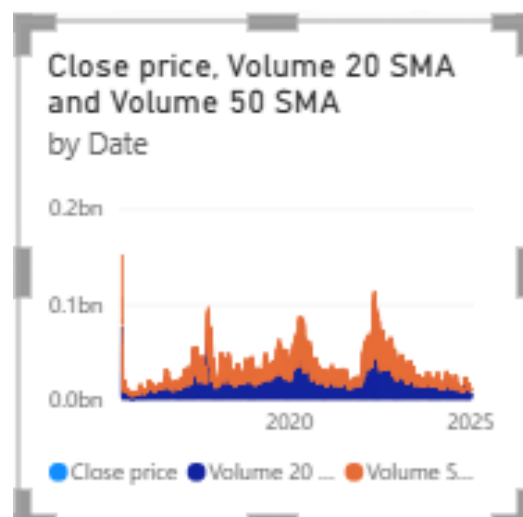
- When  $20\text{ SMA} > 50\text{ SMA}$ , the signal is displayed as Bullish ( $20 > 50$ ).
- When  $20\text{ SMA} < 50\text{ SMA}$ , the signal is displayed as Bearish ( $20 < 50$ ).

This signal helps to display the relationship between the two moving averages.

## 7. Visualizations Created

### 7.1 Stock Price and Moving Average Chart

A Line and Stacked Column Chart was created to display the stock price and moving averages over time. The Date field is used on the X-axis, while Close Price, 20 SMA and 50 SMA are used as values. A continuous date axis and zoom slider were also configured.



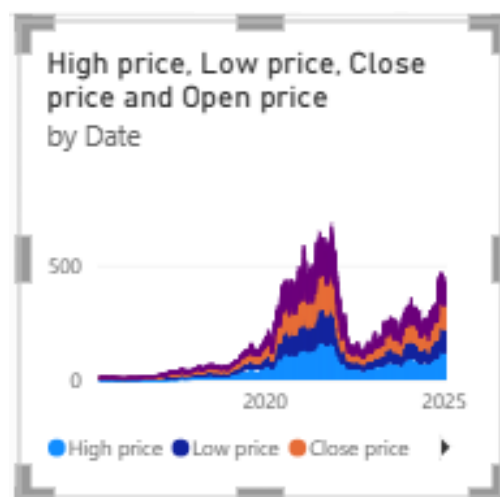
### 7.2 Trading Volume Analysis

A volume chart was created to analyse trading volume over time. Trading Volume and 20-Day SMA are used to understand changes in trading activity.



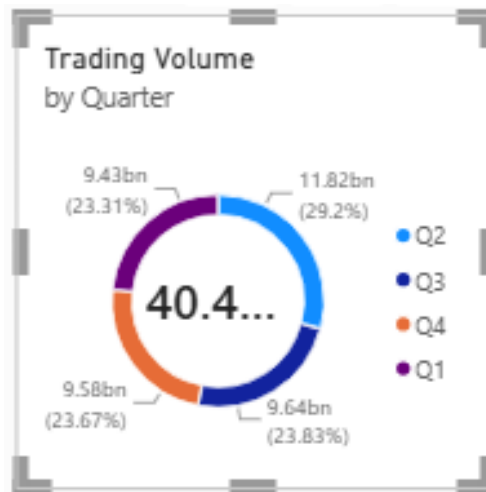
## 7.3 Open Price vs Close Price

A comparison chart was created using Open Price and Close Price to observe their movement over time.



## 7.4 Quarter-wise Trading Volume

A chart was created to compare trading volume across different quarters.



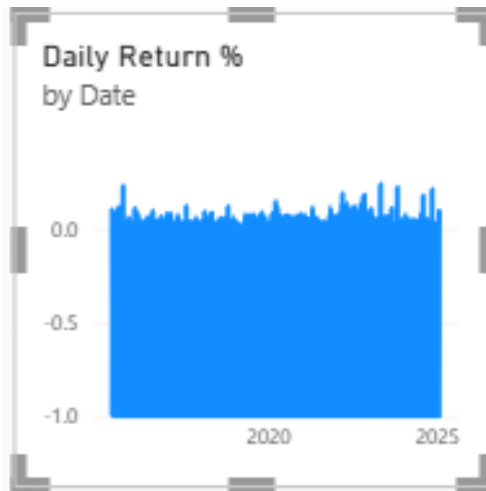
## 7.5 Year-wise Trading Volume

Year-wise trading volume was visualized to compare trading activity across different years.



## 7.6 Daily Return % Analysis

The Daily Return % chart shows the daily changes in Shopify stock price. It helps to understand the daily performance of the stock.



## 7.7 High Price and Low Price

This chart compares the highest and lowest stock prices over time. It helps to understand the price range.



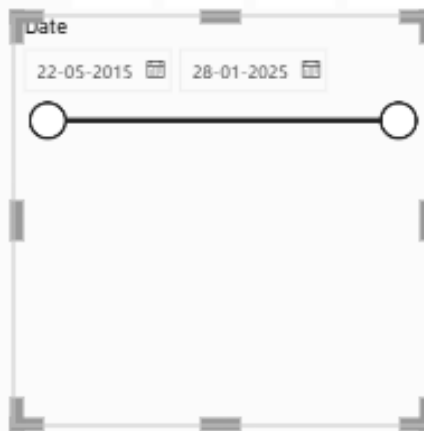
## 7.8 Moving Average Crossover Signal

The crossover signal compares the 20-Day SMA and 50-Day SMA. It displays whether the signal is Bullish or Bearish.



## 7.9 Date Range Slicer

A Date Range Slicer was added to select a specific time period. It helps users view the data for the selected dates.



## 8. Interactive Features

### Date Range Slice

A date range slicer was added to allow users to select a specific period and analyse the corresponding stock data.

## Stock Metric Selector

A field parameter was created to switch between:

- Close Price
- High Price
- Low Price
- Open Price

This makes the dashboard more interactive.

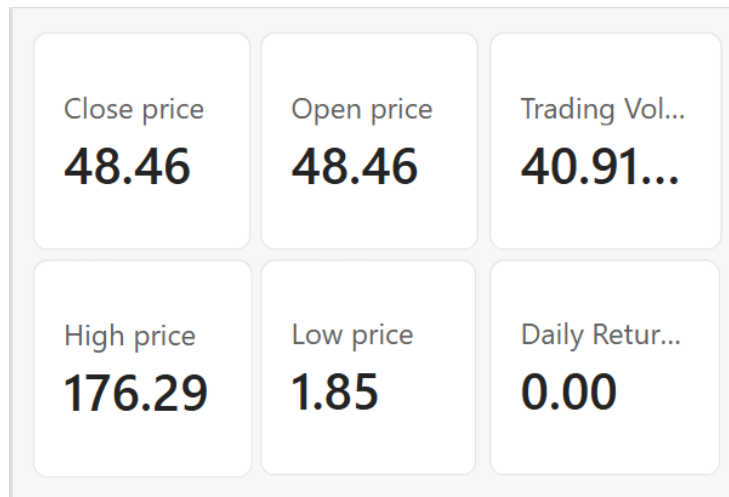
## 9. Custom Tooltip

A separate **Stock\_Tooltip** page was created as a custom tooltip.

The tooltip displays:

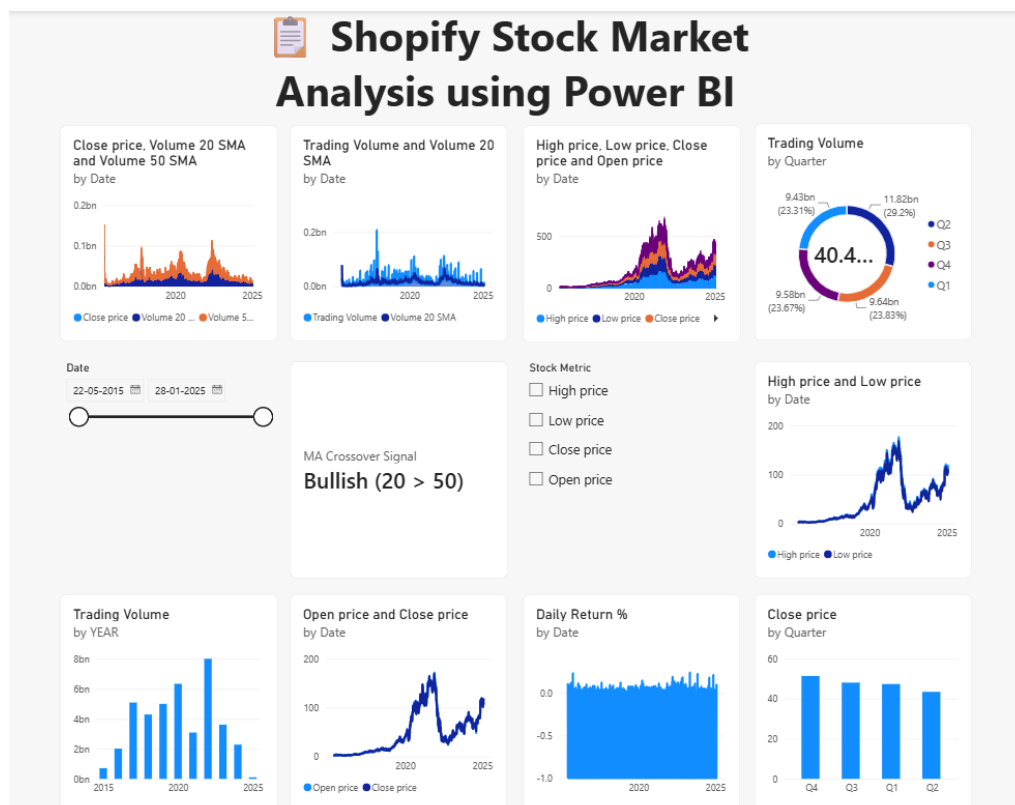
- Open Price
- High Price
- Low Price
- Close Price
- Trading Volume
- Daily Return %

The tooltip page was configured as a tooltip and connected to the main stock chart.



## 10. Dashboard

The final Power BI dashboard combines stock price trends, moving averages, trading volume, interactive slicers, metric selection and custom tooltip features in a single report page.



## 11. Key Observations

- The stock price changes over different dates.
- Trading volume varies across different time periods.
- The 20-day and 50-day moving averages provide a comparison of recent and longer-period volume trends.
- The dashboard allows users to interact with different stock metrics.
- The custom tooltip provides additional stock information when viewing the chart.

## 12. Conclusion

The Shopify Stock Market Analysis project demonstrates how Power BI can be used for Exploratory Data Analysis of stock market data. Moving averages, crossover signals, interactive visualizations, slicers and custom tooltips were used to make the analysis more informative and interactive. This project provides a clear view of stock price and trading volume trends using Power BI.